MedGuard - Compression and Optimization Results

Student: Alaa Hosny | ID: 220717702

Course: SE 4003

Date: 30 October 2026

Objective

Apply model compression and inference optimization to MedGuard, and complete the highest-priority item from last semester's future work: span-consistent training (align Stage 2/3 training with the demo's span-mean pooling). Report the measured trade-off between efficiency (size, latency, memory) and prediction quality.

Procedure

1. Establish a serving-path baseline (CLS pooling vs span-mean pooling).
2. Train a frozen-trunk Stage 3 checkpoint (shared layers frozen; severity head only).
3. Evaluate single-encoder inference (one Bio_ClinicalBERT forward).
4. Apply dynamic INT8 post-training quantization.
5. Retrain Stage 2 and Stage 3 with span-mean pooling for train/serve consistency.

Results

Serving baseline

Pooling	Interaction macro-F1
CLS	0.4190
Span-mean	0.2103

Two-checkpoint serving cost (CPU): 845.4 MB disk, 0.243 s/pair, ~1388 MB RAM.

Frozen-trunk Stage 3

Severity macro-F1 = 0.3214, within the pre-defined acceptance interval (0.2915-0.4271). Saved as stage3_severity_frozen_trunk.pt.

Single-encoder inference

	Two forwards	One forward
Latency	0.243 s	0.121 s
RAM	~1388 MB	~1153 MB

Latency improved; interaction quality under span pooling did not meet the deployment threshold.

Dynamic INT8 (PTQ)

	FP32	INT8
Disk	422.7 MB	171.0 MB
Latency	0.132 s	0.066 s
Severity span macro-F1	0.3173	0.1647
Caution / warning / danger F1	-	0

INT8 met the size and latency targets. Severity class quality did not. The quantized checkpoint is retained for the record (stage3_severity_frozen_trunk_int8.pt). Serving remains FP32.

Span-consistent retraining

Evaluation restricted to pairs absent from training:

Metric	Before	After
Interaction accuracy	0.24	0.81

Interaction macro-F1	0.2024	0.5098
Severity macro-F1	0.3285	0.4421
Danger recall	0.15	0.50

Checkpoints: stage2_interaction_span.pt, stage3_severity_span.pt (previous checkpoints retained).

Current serving configuration

Head	Checkpoint
Interaction	stage2_interaction_span.pt
NER + severity	stage3_severity_span.pt

Precision: FP32. Encoder forwards: 2.

Summary

Compression and inference optimization were implemented and measured. Dynamic INT8 reduced size and latency but did not preserve severity quality under PTQ; FP32 serving was therefore retained. Single-forward inference improved cost but did not satisfy the interaction quality requirement. Span-consistent retraining aligned training with the serving representation and improved held-out interaction accuracy (0.24 to 0.81) and danger recall (0.15 to 0.50). The academic multi-task architecture is unchanged.

Future work

Quantization-aware training for INT8 severity; expanded DDInter coverage; KG/Lipinski ablation; simplified architecture figures.